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Review

Self-regulation for responsible banking and ESG disclosure scores: Is there a link?

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A R T I C L E I N F O

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A B S T R A C T

Banks play a crucial role in sustainable development, an area increasingly governed by self-regulation. This study examines whether banks that commit to self-regulation by adopting the Principles for Responsible Banking (PRB) exhibit enhanced Environment, Social, and Governance (ESG) performance. Utilizing Bloomberg ratings, we find that PRB adopters consistently show higher ESG scores than non-adopters, both before and after adoption. This suggests that a commitment to self-regulation serves as a reliable signal of responsible banking practices. Notably, the superior performance of PRB adopters relative to non-adopters is primarily driven by a strong pre-adoption commitment to transparency on ESG issues. We discuss possible explanations for this trend, including the role of early adopters in advancing industry-wide standards. Additionally, our findings reveal a negative association between regulatory quality and ESG scores, implying that banks may leverage ESG disclosures to mitigate information asymmetries in weaker institutional environments.

1. Introduction

In response to the March 2023 US bank failures, Elizabeth Warren, a senior US senator and former law professor at several top universities, was interviewed on CNBC regarding the claim that bank-run stress tests could replace those conducted by the Federal Reserve. Warren's response was emphatic: *I'm sorry, I taught school for many, many years, and I did not let my students do their own testing.*[1] This study examines the banking industry's self-imposed commitment to principles for responsible banking and investigates the association between self-regulation in responsible banking, and transparency on Environmental, Social and Governance (ESG) issues.

Banks have the potential to become crucial players in sustainable development by influencing the allocation of funds towards investments in health, education, entrepreneurship, and poverty reduction (Demirgüç-Kunt and Levine, 2008; Contreras et al., 2019; Bhutta et al., 2022; Úbeda et al., 2022b; Grijavo and García-Wang, 2023; Úbeda et al., 2023). This raises the question of how best to encourage banks to engage in sustainable development. A common response is that sustainable development is most effectively achieved through public–private cooperation and self-regulation, rather than solely relying on government-imposed rules (Pitkänen et al., 2016; Folke et al., 2019; Kolcava et al., 2021). Consequently, many authors stress the importance of inter-firm collaborations,

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[1] https://twitter.com/atrupar/status/1636014248990998531?ref_src=twsrc%5Etfw%7Ctwcamp%5Etweetembed%7Ctwterm%5E1638172644435542016%7Ctwgr%5Eb048d07c083efef45deafdafd216337946c365b6%7Ctwcon%5Es3_&ref_url=https%3A%2F%2Fwww.the-marker.com%2Fwallstreet%2F2023-03-22%2Fty-article%2F.premium%2F00000187-09f5-d1cf-a7af-fdfd4ed90000.

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voluntary sectoral self-regulation, and private–public co-regulation as effective tools for managing sustainability (e.g., Anton et al., 2004; Arimura et al., 2008; Contreras et al., 2019; Kolcava et al., 2021).

Within the banking industry, several studies investigate the phenomenon of self-regulation in sustainability finance (see, for example, Wright and Rwabizambuga, 2006; Scholtens and Dam, 2007; O'Sullivan and O'Dwyer, 2009; Macve and Chen, 2010). Most of these studies, however, focus on the *Equator Principles* (EP), which were formally launched in 2003 to guide banks in determining, assessing, and managing environmental and social risks in *project finance* (Abudy et al., 2023).[2] Notwithstanding these studies and given the central role of banks in advancing sustainability, Contreras et al. (2019) highlight the lack of empirical evidence in this area. This study aims to address the gap by assessing whether banks that voluntarily commit to the discipline of a comprehensive set of responsible banking principles – specifically, signatories to the *Principles for Responsible Banking* (PRB) − tend to score higher on ESG disclosure.

The PRB were launched in 2019 through a partnership between funding banks and the United Nations (UN). They include six principles that aim to align signatories' strategy and practices with the *Sustainable Development Goals* (SDG[3]) and the *Paris Climate Agreement*.[4] Unlike the EP, which focus on project finance in developing countries, the PRB offer a *comprehensive framework* for sustainable finance and responsible banking "across *all* business areas, at the strategic, portfolio, and transactional levels".[5] Regardless of their differences, however, both the EP and the PRB are part of a greater trend towards soft law and self-regulation by industry and corporations for dealing with grand environmental and social challenges (e.g., Anton et al., 2004; Graham and Woods, 2006; Barla, 2007; Albareda, 2008; Chen and Huang, 2023).

This trend towards self-regulation as a corporate tool to achieve corporate responsibility and sustainability is linked to the debate on the appropriate balance between government-imposed regulation, self-regulation, and public–private collaborations (e.g., Barla, 2007; Arimura et al., 2008; Pitkanen et al., 2016; Contreras et al., 2019; Folke et al., 2019; Aragon-Correa et al., 2020).[6] In the banking industry, this has led to numerous studies investigating various aspects of self-regulation for responsible banking (e.g., Scholtens and Dam, 2007; Finger et al., 2018; Contreras et al., 2019). For example, Scholtens and Dam (2007) conclude that self-regulation requires banks to adhere to stricter rules than those required by law and can thus act as a signal for responsible banking. This study contributes to the debate by examining whether voluntarily becoming a PRB signatory and adhering to the principles' self-imposed guidelines is associated with higher ESG disclosure scores assigned by third-party raters.

ESG disclosure scores have gained significant public and corporate attention as mounting environmental and social challenges have prompted stakeholders to demand greater transparency and improved performance on these issues (e.g., Eccles et al., 2011; Siew, 2015; Yu and Van Luu, 2021; Fernhaber and Zou, 2022). A popular tool used by stakeholders to assess corporations' progress toward sustainability goals is Bloomberg's ESG disclosure score (Siew, 2015),[7] which is viewed as an investment in sustainability. This investment contributes to financial stability and may help to re-establish trust between banks and their investors, particularly in the aftermath of shocks such as the 2008 global financial crisis (Trinh et al., 2023; Gupta and Kashiramka, 2024). The popularity of this and other disclosure ratings may be attributed to empirical evidence showing that ESG disclosure can provide valuation benefits for firms (e.g., Orlitzky and Benjamin, 2001; Bansal and Clelland, 2004; Gracia et al., 2017; Azmi et al., 2021).

The question we ask is whether banks that commit to self-regulation by adopting principles for responsible banking (i.e., signing the PRB) tend to score higher on Bloomberg's ESG disclosure. In line with Scholtens and Dam (2007), we hypothesize that PRB adoption serves as a signal of quality and characterizes banks that are fully engaged and act as industry leaders in responsible conduct. Alternatively, self-regulation is sometimes viewed as a form of greenwashing, where the public is misled about an organization's environmental performance (see Delmas and Burbano, 2011; Caby et al., 2020). In this study, we empirically investigate whether PRB membership is positively associated with improved performance on ESG transparency, as indicated by ESG disclosure scores assigned to banks by Bloomberg, a third-party ESG disclosure rater.

Notwithstanding our aim, when assessing the association between self-regulation and ESG disclosure performance, the regulatory context must not be overlooked, as it has been shown to explain many cross-country variations in corporate behavior (La Porta et al., 1998; Dixon-Fowler et al., 2013; Breuer et al., 2018; Miralles-Quirós et al., 2019; Erragragui et al., 2023). Amongst other factors, weak formal institutions and a fragile regulatory environment form barriers to achieving sustainable development (Karim et al., 2022; Úbeda et al., 2022a, 2022b). Such conditions create uncertainty and risk, which firms can address by demonstrating responsible conduct and enhancing ESG-related disclosure, thereby establishing trust (Egginton and McBrayer, 2019; Huang et al., 2022; Úbeda et al., 2022a, 2022b) and creating value (Azmi et al., 2021). In line with this reasoning, we argue that banks operating in weak regulatory environments leverage ESG disclosure to mitigate information problems.

Our sample spans the period from 2016 to 2022 and includes 84 publicly traded banks that adopted the PRB, along with a matched sample of non-adopters. The dependent variable is ESG disclosure, measured using Bloomberg's ESG disclosure scores, with the change in score used as an alternative dependent variable. The main explanatory variable of interest is a dummy variable for PRB adoption,

[2] https://equator-principles.com/.
[3] https://sdgs.un.org/.
[4] https://www.unepfi.org/banking/bankingprinciples/.
[5] https://www.unepfi.org/banking/more-about-the-principles/.
[6] Self-regulation is often referred to as private regulation, while co-regulation refers to collaboration between the private and public sectors in designing and enforcing regulations.
[7] Bloomberg official website: https://bloomberg.com.

while control variables consist of firm-level factors and Kaufmann et al.'s (2011) regulatory quality indicator.[8]

Initial findings from comparative analyses reveal that PRB adopters tend to have higher ESG scores, demonstrating a stronger commitment to responsible conduct compared to non-adopters. Interestingly, this commitment to improving ESG scores diminishes somewhat after adoption, relative to pre-adoption levels. In contrast, non-adopters show greater improvements in their ESG scores during the post-PRB period compared to the pre-PRB period. Consequently, the changes in ESG scores for both adopters and non-adopters are not significantly different in the post-PRB period.

Further regression analyses, including mixed-effects regressions, propensity score matching, and difference-in-difference procedures, confirm these findings. The results affirm that PRB adopters have higher ESG scores than non-adopters in both the pre- and post-PRB adoption periods. This is largely because PRB adopters demonstrate a strong commitment to improving their ESG scores, particularly in the early period (pre-PRB adoption). Consistent with Johannsdottir and McInerney (2018), we interpret these findings to suggest that PRB signatories are the industry's early adopters of responsible banking, thereby driving the broader adoption of sustainability practices. In this way, leaders' examples and peer pressure increase both the speed and scale of industry-wide implementation of sustainable banking practices (see Contreras et al., 2019).

Additionally, our analysis reveals that banks from countries with low regulatory quality tend to have higher ESG scores compared to banks from other countries. This suggests that ESG disclosure helps mitigate information asymmetry and uncertainty in weak institutional environments. We also find that being an EP signatory generally has a positive impact on ESG disclosure scores. Lastly, while this relationship is nonlinear, our findings indicate that PRB adopters tend to be the largest banks, supporting the view that industry leaders are often early adopters (Johannsdottir and McInerney, 2018).

The paper contributes to existing literature in four ways. First, we add empirical evidence to the debate regarding the role of self-regulation in sustainable management, in line with studies such as Blacconiere and Patten (1994), Graham and Woods (2006), Sekarlangit and Wardhani (2021), Chen and Huang (2023), among others. Second, our paper contributes to the literature on the value of self-regulation in responsible banking by examining the PRB, given that most other studies focus on the more narrowly defined EP (e.g., Wright and Rwabizambuga, 2006; Scholtens and Dam, 2007; Macve and Chen, 2010). Third, the paper adds to existing work (e.g., Wu, and Shen, 2013; Finger et al., 2018; Caby et al., 2020) related to greenwashing versus signaling in the banking industry. Fourth, the paper contributes to the literature on sustainable banking in weak institutional environments by providing evidence on the association between ESG disclosure and the quality of the regulatory environment (e.g., Azmi et al., 2021; Gupta and Kashiramka, 2024).

The remainder of the paper is organized as follows. The next section provides background on self-regulation in responsible banking and on the PRB. Section 3 reviews relevant literature and presents the hypothesis. In section 4, we describe the methods and data, while section 5 presents and discusses the results. Section 6 concludes the study.

2. The principles for responsible banking (PRB)

Until 2019, the main self-regulation for responsible banking was the Equator Principles (EP) initiative, which dates back to 2003. The EP framework consists of ten best practice principles designed to guide financial institutions in identifying, assessing, and managing environmental and social risks when providing credit for large infrastructure and industrial projects in developing countries (i.e., non-designated countries).[9]

This initiative originated from ABN AMRO's experience in financing a mining project in Papua New Guinea that severely contaminated local water supplies. Motivated by this, the bank approached the International Finance Corporation (IFC) and three other industry leaders – Barclays, Citigroup, and WestLB – to discuss potential solutions to such problems (Scholtens and Dam, 2007). Indeed, research generally confirms that the EP framework has yielded positive outcomes (Scholtens and Dam, 2007; Macve and Chen, 2010; Finger et al., 2018; Contreras et al., 2019).

However, in 2019, the UN Environment Programme Finance Initiative (UNEPFI[10]) initiated the PRB. Like the EP, the PRB initiative is specific to the banking sector and guides signatories on how to consider the social and environmental impact of their banking decisions. The PRB is designed to help banks align their business strategies with the SDG and the Paris Climate Agreement, two programs that were approved by the UN in 2015. The UNEPFI describes the principles as "…*the world's foremost sustainable banking framework*".[11] They consist of six principles to help in the implementation of a range of sustainable practices, including setting targets, reporting on progress, engaging with stakeholders, and conducting impact assessments.

Prominently, the PRB framework is designed to ensure that sustainability is integrated into *all aspects* of banking activities. This can be contrasted with the EP framework which focuses on *a single* aspect of banking, namely project finance. Similarly, in terms of size (as of the end of 2023), the two initiatives differ, with 534 PRB adopters versus 136 EP adopters. Certainly, the rapid PRB adoption rate may reflect the growing number of senior executives and decision-makers who expect the benefits of PRB adoption to exceed the costs associated with implementing it, reinforcing the perceived value and role of this self-regulation initiative (see Lyon and Maxwell (2003) on self-regulation theory). Moreover, the popularity of the PRB is also indicative of the rapidly expanding consensus on the importance of sustainable management (Mitchell, 2017; Rizzi et al., 2018).

[8] WGI official website: https://info.worldbank.org/governance/wgi/.

[9] https://equator-principles.com/about-the-equator-principles/.

[10] United Nations Environment Programme Finance Initiative (UNEPFI) official website: https://www.unepfi.org/.

[11] From the official PRB website: https://www.unepfi.org/banking/bankingprinciples/.

3. Literature review and hypothesis development

3.1. Hypothesis development

Sustainable finance, characterized by investments that consider ESG factors, has gained prominence as a critical element in addressing global challenges such as climate change, financial exclusion, and social inequality (Taneja and Ali, 2021; Úbeda et al., 2022a, 2022b). More specifically, sustainable banking – designing products and services that consider social and environmental objectives while generating profits – is a key element in achieving the SDG, due to the crucial role of banks in providing capital and mobilizing financial resources, including towards ESG projects (Caby et al., 2020; Aracil et al., 2021). Thus, the widespread acceptance of sustainable banking is crucial for achieving economic and social well-being (e.g., Caby et al., 2020).

Consequently, recent years have seen increasing number of banks getting involved in ESG and sustainability, while stakeholders – including investors and customers – are paying growing attention to whether banks behave in a responsible manner (Taneja and Ali, 2021; Agnese and Giacomini, 2023). In particular, interest in sustainable banking has gained momentum following the global financial crisis of 2008, driven by the industry's attempt to recover its damaged reputation (Aracil et al., 2021; Úbeda et al., 2022a, 2022b; Trinh et al., 2023; Gupta and Kashiramka, 2024).

In response to this trend, self-regulation has emerged as a key voluntary mechanism for companies to manage their ESG risks and opportunities, and to demonstrate their commitment to sustainable business practices (Lyon and Maxwell, 2003; Anton et al., 2004; Chen and Huang, 2023). This has led to debates and criticisms regarding the phenomenon of self-regulation on environmental and social issues (e.g., Denicolo, 2008; Sethi and Schepers, 2014; Malhotra et al., 2019). For example, Boiral and Heras-Saizarbitoria (2017) argue that companies gain and reinforce social acceptability and legitimacy by the implementation of voluntary standards and guidelines related to environmental protection. Malhotra et al. (2019) claim that firms use private regulations as a political strategy to pre-empt more stringent public regulations.

Within the context of sustainable finance, self-regulation refers to voluntary initiatives and internal governance mechanisms adopted by financial institutions to align their operations with the SDG. Such commitment is said to constitute a mechanism to build trust and create opportunities for advancing sustainable banking and addressing social and environmental challenges (Úbeda et al., 2022b). Other research has resorted to two alternative theories to explain why firms voluntarily commit to sustainable reporting, namely signaling theory (Spence, 1973) and greenwashing theory (Mahoney et al., 2013; Caby et al., 2020; Uyar et al., 2020). Signaling theory suggests that the voluntary adoption of self-regulation for responsible conduct is a signal of commitment to sustainability. Greenwashing theory suggests that such actions aim to position the firm as sustainable and responsible, even when it does not necessarily conduct itself in that way, damaging consumer confidence in green products and investor confidence in environmentally friendly firms (Delmas and Burbano, 2011).

For example, Uyar et al. (2020) find that companies in the logistics sector are more likely to report their corporate social responsibility (CSR) performance when it is high. These results are in line with the signaling theory and inconsistent with greenwashing theory. Finger et al. (2018) find that EP adoption is a strategic decision for banks in developing countries. It requires commitment and investment to create value; hence, in line with signaling theory, the share price reaction to EP adoption is positive. In contrast, in developed countries, EP adoption does not require extra commitment and is therefore no more than an attempt at greenwashing (Finger et al., 2018).

Criticism aside, banks the world over have been signing up for self-imposed regulations and committing to sustainability management practices. This trend may reflect the general view regarding the responsibility banks have towards their customers, affected sections of society, and the environment in general (e.g., Kara et al., 2022). Several studies seek to understand what characterizes banks that adopt codes of conduct or self-regulation related to sustainability, as well as to shed light on whether these actions amount to genuine commitment to responsible conduct or merely a form of greenwashing (e.g., Scholtens and Dam, 2007; Finger et al., 2018; Finger and Rosenboim, 2022).

Scholtens and Dam (2007) find that the ESG policies of banks that adopted the EP differ significantly from those of non-adopters. They conclude that signing up for the EP serves as a signal of responsible conduct. Contreras et al. (2019) argue that banks' commitment to self-regulation in sustainable management can be explained by their exposure to peer pressure. Their study finds that banks cooperating with EP signatories tend to join this voluntary self-regulation as well. Brandon et al. (2022) study the investment industry and the impact of adopting the Principles for Responsible Investment (PRI). Founded in 2006 through cooperation between the UN and the world's largest institutional investors, these principles provide a voluntary framework to guide signatory investors in incorporating ESG issues into their investment processes.[12] Brandon et al. (2022) find that non-US institutional investors who signed the PRI not only showed better portfolio ESG scores but also improved these scores after adopting the principles. In line with these studies, which generally find significant differences between adopters and non-adopters of self-regulation for sustainable conduct, and consistent with signaling theory (Spence, 1973), we hypothesize as follows:

H1. *PRB signatories have higher ESG disclosure scores awarded by an external rater, compared with other banks.*

[12] Information on the PRI is available at: https://www.unpri.org/about-us/about-the-pri.

3.2. The regulatory context

The growing global trend for companies committing to sustainability and disclosing their ESG performance aligns with stakeholder theory (Freeman and McVea, 2000) and legitimacy theory (Suchman, 1995). ESG disclosure reduces information asymmetry and establishes legitimacy by meeting stakeholders' expectations (Garcia et al., 2017; Shin et al., 2023). By increasing transparency, ESG disclosure reduces uncertainty and business risk, reinforces the firm's credibility and stakeholders' trust, enhances access to capital, shields the firm from criticism, reduces its cost of equity, and builds reputational value (Boiral and Heras-Saizarbitoria, 2017; Li et al., 2018; Albarrak et al., 2019; Miralles-Quirós et al., 2019; Trinh et al., 2023; Gupta and Kashiramka, 2024). For example, using FTSE-listed firms from various sectors, Li et al. (2018) find a positive association between ESG disclosure scores and firm value. Similarly, Agnese and Giacomini (2023) find that the cost of issuing bonds is lower for banks with high ESG scores.

Nevertheless, the value of corporate commitment to sustainability and responsible management also depends on the circumstances under which firms operate. For example, McWilliams et al. (2006) argue that stakeholders' demand for CSR varies substantially across nations, regions, and industrial sectors (see also La Porta et al., 1998; Miralles-Quirós et al., 2019; Erragragui et al., 2023). Similarly, Breuer et al. (2018) examine the effects of ESG performance on firms' cost of equity under different levels of investor protection. They find that in countries with high investor protection, the cost of equity falls as ESG performance increases, while this relationship is reversed in countries with low investor protection. Zhang et al. (2021a) suggest that in the absence of adequate mandatory water legislation, voluntary disclosure of water management practices by large, listed companies in water-sensitive industries plays an important role in sustainable water management. According to Gupta and Kashiramka (2024), ESG disclosure enhances the stability of banks in economies with poor transparency and governance mechanisms. Trinh et al. (2023) show that during periods of heightened risk and uncertainty, such as post-crisis phases, banks with high ESG ratings are less vulnerable to severe market downturns.

Other studies attempt to explain the context-dependent value of ESG commitment. For example, Graham and Woods (2006) propose that voluntary social and environmental management by multinational corporations can mitigate the weak regulatory environments that often characterize developing states. Úbeda et al. (2022b) argue that sustainable banking establishes trust, which helps to overcome institutional weaknesses, such as a weak rule of law. Similarly, Garcia et al. (2017) explore the idea that corporate ESG practices are in greater demand where they matter most, such as in emerging markets or industries with a high risk of causing social and environmental damage.

Consistent with those studies, Huang et al. (2022) find that firms increase ESG disclosure when operating in perceived high-risk environments. This is in line with Egginton and McBrayer (2019), who argue that firms employ ESG disclosure to mitigate problems associated with asymmetric information. Similarly, Yu and Van Luu (2021) find that cross-listed firms use ESG disclosure to signal their quality and reduce the informational disadvantage they face relative to local firms when accessing local capital.

Following this line of thinking, we argue that in strong regulatory environments, firms have less incentive to disclose their ESG practices. In contrast, firms operating in countries with lower regulatory quality utilize ESG disclosure to signal their quality and mitigate information problems, as the value of the signal is higher under such conditions.

4. Sample, Data, and Methodology

Our sample spans the period from 2016 to 2022 and includes 84 publicly listed banks that adopted the PRB, along with a matched sample of 1,242 non-adopters. The matched sample of non-adopters includes all publicly traded banks from the 29 countries represented in the adopting sample. Each bank has seven years of observations (though not for all variables), totaling 9,282 year/bank observations.

To test H1, we define the dependent variable, *ESG*, as Bloomberg's ESG disclosure score, which is calculated based on the guidelines of the Global Reporting Initiative (GRI) and reflects disclosure in three major categories: environmental, social, and governance.[13] Disclosure performance in these categories is measured using 100 sub-categories, with their weighting determined by the Bloomberg Industry Classification System (BICS) to account for business differences (see Eccles et al., 2011; Siew, 2015; Smeesters et al., 2018). Examples of sub-categories include board characteristics, diversity and inclusion policies, electricity usage, and environmental certification. The score is updated annually and ranges from 0 (the lowest) to 100 (the highest). Bloomberg's ESG disclosure score is considered a measure of commitment to transparency and ESG practices (Eccles et al., 2011; Tamimi and Sebastianelli, 2017; Yu and Van Luu, 2021). As an alternative to using *ESG* as the dependent variable, we also use the yearly change in the ESG score, *dESG*. The one-year lagged ESG score, *Lagged ESG*, is included as a control variable in regressions where the dependent variable is *dESG*.

To distinguish between PRB adopters and non-adopters, a dummy variable, *Adopter*, is included, which takes the value of 1 if the bank is a PRB adopter and zero otherwise. Similarly, to separate the pre- and post-adoption periods, we use a dummy variable, *Adopter Post*, which takes the value of 1 for adopters in the year of adoption or afterword, and zero otherwise. To control for EP adoption, we include a dummy, *EP Adopter*, that equals 1 if the bank is an EP signatory and zero otherwise. In addition, to proxy for the country's regulatory quality, we extract data on this indicator from the Worldwide Governance Indicators (WGI) database.[14] *Reg Qlty* is a continuous variable that reflects perceptions related to the ability of a country's government to create and implement effective policies and regulations that encourage the growth of the private sector (Kaufmann et.al., 2011). Appendix Table 1 provides a list of countries with their average regulatory quality indicators.

[13] Bloomberg official website: https://bloomberg.com.

[14] WGI official website: https://info.worldbank.org/governance/wgi/.

Variables at the bank level, acting as control variables, are based on data obtained from Bloomberg. They include firm size (log of assets), and the size squared to account for a possible nonlinear relationship (*size* and *size2*, respectively), return on assets (*ROA*), and the Tier 1 ratio (*tier1*), which measures the ratio of capital to risk-weighted assets. Table 1 summarizes the definitions of all variables, while Table 2 presents descriptive statistics and correlations.

To reduce the impact of outliers, all non-dummy variables in Table 2 and beyond are winsorized at the 1st and 99th percentiles. That is, values smaller than the 1st percentile are replaced by the 1st percentile, and values greater than the 99th percentile are replaced by the 99th percentile. Looking at Table 2, we see that, relative to the full sample, PRB adopters have a higher mean ESG score, reside in countries with a higher average regulatory quality, tend to be larger in size, are less profitable, and have a higher average Tier 1 ratio.

Table 3 provides basic comparative analyses and is split into five panels. Panel 3A shows that average ESG scores generally increase from year to year, while the average yearly change in ESG is always positive but does not appear to follow a particular pattern. Panel 3B presents the yearly changes in ESG scores, split into PRB adopters and non-adopters. It shows that the average yearly changes for PRB adopters are *higher* in the pre-PRB period (up to 2019) than in the post-PRB period. No such pattern can be observed for non-adopters.

Panel 3C delves deeper into these apparent differences, comparing the ESG scores of PRB adopters and non-adopters over the entire period, as well as in the pre- and post-PRB periods. As shown, the average ESG scores of PRB adopters are always significantly higher than those of non-adopters. This is true for the entire period (49.62 for PRB adopters versus 33.58 for non-adopters), the pre-PRB period (48.28 for PRB adopters versus 32.02 for non-adopters), and the post-PRB period (50.79 for PRB adopters versus 34.73 for non-adopters). Panel 3D shows that the difference between PRB adopters and non-adopters in the yearly change in ESG scores during the post-PRB period (1.09 for PRB adopters versus 0.93 for non-adopters) is not significant. This contrasts with the equivalent comparison in the pre-PRB period (2.01 for PRB adopters versus 0.29 for non-adopters, which is a significant difference).

Panel 3E presents comparisons of the average yearly change in ESG scores between the pre- and post-PRB periods, separately for PRB adopters and non-adopters. A stark difference between the two groups is evident. Specifically, for PRB adopters, the average yearly increase in ESG scores is significantly higher in the pre-PRB period (1.97) than in the post-PRB period (1.11). Conversely, for non-adopters, the average yearly increase in ESG scores is significantly lower in the pre-PRB period (0.29) than in the post-PRB period (0.93).

Together, the results in Table 3 indicate that ESG scores increase over time for both PRB adopters and non-adopters, although the average score remains consistently higher for PRB adopters. Additionally, while the yearly increases in ESG scores are always positive, the trend differs between PRB adopters and non-adopters. PRB adopters appear to have recognized the importance of ESG disclosure earlier and were the first to take actions that improved their ESG scores. Non-adopters began to catch up only in the later period, at which point the difference in the average yearly change in ESG scores between the two groups narrowed from 1.72 in the pre-PRB period to 0.16 in the post-PRB period, becoming statistically insignificant. These results only partially confirm H1, aligning with the idea that PRB adoption was a reliable signal of commitment to responsible banking only in the pre-PRB period.

The next stage of the empirical procedure involves regression analyses. It begins with mixed-effect regressions of ESG disclosure scores (panel 4A) and the changes in ESG disclosure scores (panel 4B) on the explanatory variables over the entire period (2016–2022) and over the post-PRB period (2019–2022). However, evaluating the effects of PBR adoption by comparing adopting banks with non-adopters is problematic. The reason is that we do not know what the ESG disclosure scores of non-adopters would have been had they decided to adopt the PRB (see Zhang et al., 2021b). Rosenbaum and Rubin (1983) propose propensity score matching (PSM) to address this problem by finding non-participants with the same characteristics as participants. Thus, in the second stage of the empirical procedure, we follow Bazel-Shoham et al. (2023) and use the PSM procedure to further strengthen our results (Tables 5 and 6). Another possible approach to comparing ESG scores over time between PRB adopters and non-adopters is the difference-in-differences approach (Wing et al., 2018), which we also perform and report in Table 7.

Table 1
Definitions of variables.

Variable	Description
ESG	Bloomberg's ESG disclosure scores reflect the bank's level of disclosure on environmental, social, and governance issues. The score ranges from 0 to100, where 0 indicates poor disclosure and 100 indicates high disclosure.
Lagged ESG	The previous year's ESG disclosure score.
dESG	The yearly change in ESG.
Adopter	A dummy variable set to 1 if the bank signed the PRBs and 0 otherwise, based on information on the PRB official website.
Adopter Post	A dummy variable set to 1 if the bank is a PRB adopter and the year is the year of adoption or later, and 0 otherwise.
EP Adopter	A dummy variable set to 1 if the bank is a member of the Equator Principles, and 0 otherwise.
size	The natural logarithm of the bank's total assets.
size2	The variable size, squared.
ROA	Return on assets is a measure of profitability, defined as: (Net Income / Average Total Assets) * 100.
tier1	Tier 1 Ratio of capital to risk-weighted assets.
Reg Qlty	Regulatory Quality: Reflects perceptions of the government's ability to formulate and implement sound policies and regulations that permit and promote private sector development. The score ranges from −2.5 (weak) to 2.5 (strong) governance performance (Kaufmann et.al., 2011).

Table 2
Descriptive Statistics and Correlations.

Panel 2A: Descriptive statistics for non-dummy variables

	Full Sample				Sample of 84 PRBs adopters			
Variable	Mean	Std. dev.	Min	Max	Mean	Std. dev.	Min	Max
ESG	35.583	10.463	14.244	63.082	49.619	10.695	18.207	63.082
dESG	0.790	3.040	−12.846	12.112	1.454	3.313	−12.846	12.112
size	9.062	2.194	4.857	14.672	11.847	2.117	5.369	14.672
size2	86.942	42.606	23.587	215.263	144.818	47.460	28.828	215.263
ROA	0.876	0.704	−1.371	4.133	0.761	0.703	−1.371	4.133
tier1	14.101	4.064	7.060	30.880	15.654	3.306	8.170	28.700
Reg Qlty	1.199	0.642	−0.908	1.894	1.016	0.826	−0.908	1.894

Panel 2B: Correlation Matrix

	ESG	dESG	size	size2	ROA	tier1	Reg Qlty
ESG	1.000						
dESG	0.249	1.000					
size	0.748	0.143	1.000				
size2	0.760	0.137	0.994	1.000			
ROA	−0.246	−0.068	−0.242	−0.260	1.000		
tier1	0.100	0.039	0.003	0.011	0.082	1.000	
Reg Qlty	−0.320	−0.081	−0.380	−0.373	−0.032	0.122	1.000

Notes: A balanced panel of 1,326 banks from 29 countries over 7 years from 2016 to 2022. Variables are defined in Table 1. All non-dummy variables were winsorized at the 1st and 99th percentiles. That is, values smaller than the 1st percentile are replaced by the 1st percentile, and values greater than the 99th percentile are replaced by the 99th percentile.

5. Results

Table 4 presents the mixed-effect regression results of ESG disclosure scores and their yearly changes on the explanatory variables, incorporating bank random effects and robust standard errors.

In panel 4A, the dependent variable is the ESG disclosure score. Supporting H1, the PRB-adopter dummy (*Adopter*) is positive and highly significant, regardless of the inclusion or exclusion of control variables or year fixed effects. This indicates that PRB signatories have significantly higher ESG disclosure scores, as awarded by an external evaluator (Bloomberg), compared to non-signatories. We see that the ESG scores of adopters are even higher in the post-PRB period, as reflected by the significantly positive estimated coefficients on *Adopter Post* (columns (2)-(3) of panel 4A) and on *Adopter* (column (5) of panel 4A), where the regression is run in the post-PRB period only.

The regulation quality variable (*Reg Qlty*) is significantly negative, supporting the view that ESG disclosure is used to mitigate information problems that arise in weak regulatory environments. Most of the control variables are also significant. EP signatories and banks with lower profitability or a high ratio of capital to risk-weighted assets tend to have higher ESG disclosure. The relationship between bank size and ESG scores is nonlinear, as size has a negative effect on ESG scores up to a certain point, after which this relationship reverses. This finding is consistent with Tamimi and Sebastianelli (2017), who report that large-cap companies have significantly higher ESG scores. They explain this in terms of larger firms having the capacity to allocate resources toward both ESG practices and disclosure (see also Gianfrate, 2017). Giannarakis (2014) presents similar results and argues that the greater visibility of large companies attracts external stakeholders' attention, leading these firms to disclose more information thereby gaining legitimization.

Panel 4B shows that PRB adoption is associated with larger changes in ESG scores (*dESG*), indicating that adopters are more committed to improving their yearly ESG disclosure compared to non-adopters. However, the significantly negative estimated coefficient on *Adopter Post* (columns (3)-(6) of panel 4B) suggests that this effect weakens in the post-PRB period, though it does not disappear entirely, as evidenced by the smaller absolute coefficient on *Adopter Post* compared to *Adopter*. These findings align with the comparative analysis in Table 3, where PRB adopters exhibit significantly higher average yearly changes in ESG scores relative to non-adopters in the pre-PRB period – a difference that becomes insignificant in the post-PRB period (panel 3D). Johannsdottir and McInerney (2018) argue that smaller companies often adopt sustainability practices later, learning from early adopters. This supports our findings, as PRB adopters – typically larger banks (Table 2, panel 2A) – are the industry's leaders and early adopters in responsible banking, fostering wider adoption of sustainability practices. Consequently, industry leaders and peer influence encourage non-PRB adopters to improve their ESG scores, enhancing both the speed and scale of industry-wide ESG disclosure (see Contreras et al., 2019).

The significantly negative coefficient on the lagged ESG score (*Lagged ESG*) implies that banks with lower ESG scores in one year strive to see their scores improve in the following year. The nonlinearity observed in the relationship between ESG scores and size generally disappears when *ESG* is replaced by *dESG*, as does the effect of being an EP adopter. However, the rest of the control variables maintain their significance and signs, including *Reg Qlty*, which remains highly significant and negative.

Notwithstanding the results thus far, Rosenbaum and Rubin (1983: 41) claim that "*inferences about the effects of treatments involve speculations about the effect one treatment would have had on a unit which, in fact, received some other treatment*". To address this issue, we follow the existing literature (e.g., Dhaliwal et al., 2016; Bazel-Shoham et al., 2023) and use the PSM method to re-test our hypothesis.

Table 3
Comparison Analysis of the ESG disclosure scores.

Panel 3A: Trend in the yearly average ESG scores (*ESG*) and changes thereof (*dESG*) from 2016 to 2022 – all banks

Year	*ESG* – Mean	*ESG* – Std Err	*dESG* – Mean	*dESG* – Std Err
2016	32.969	0.433		
2017	34.419	0.433	1.460	0.109
2018	35.181	0.429	1.128	0.091
2019	36.004	0.430	0.993	0.099
2020	36.792	0.441	0.995	0.091
2021	37.376	0.444	1.158	0.111
2022	35.934	0.492	0.444	0.105

Panel 3B: Trend in the average yearly change in ESG scores (d*ESG*) from 2016 to 2022 – Adopters vs. Non-adopters

	PRB adopters		*Non-Adopters*	
Year	*dESG* – Mean	*dESG* – Std Err	*dESG* – Mean	*dESG* – Std Err
2017	1.804	0.302	1.406	0.117
2018	1.835	0.303	1.018	0.093
2019	1.350	0.369	0.935	0.099
2020	0.988	0.254	0.996	0.097
2021	1.088	0.317	1.169	0.118
2022	0.518	0.674	0.439	0.104

Panel 3C: Comparison of ESG scores (*ESG*) – PRB adopters vs. non-adopters

	Entire period (2016–2022)			Pre-PRB period (2016–2018)			Post-PRB period (2019–2022)		
	Obs.	Mean	Std Err	Obs.	Mean	Std Err	Obs.	Mean	Std Err
Non-adopters	3,332	33.58	0.15	1,419	32.02	0.21	1,913	34.73	0.21
PRB Adopters	476	49.62	0.49	222	48.28	0.72	254	50.79	0.66
Combined	3,808	35.58	0.17	1,641	34.22	0.25	2,167	36.61	0.23
Diff		−16.04	0.44		−16.26	0.61		−16.06	0.62
t (d.f.)		−36.30	(3,806)		−26.55	(1,639)		−26.02	(2,165)
Ha: *Diff*<0	Prob:	0.000		Prob:	0.000		Prob:	0.000	
Ha: *Diff*>0	Prob:	1.000		Prob:	1.000		Prob:	1.000	

Panel 3D: The change in ESG scores – PRB adopters vs. non-adopters

	dESG Entire period (2016–22)			*dESG2* Entire period (2016–22)			*dESG* Pre-PRB (2016–18)			*dESG* Post-PRB (2019–22)		
	Obs.	Mean	Std Err	Obs.	Mean	Std Err	Obs.	Mean	Std Err	Obs.	Mean	Std Err
Non-adopters	2,930	0.70	0.06	2,217	2.03	0.08	1,087	0.29	0.12	1,843	0.93	0.05
PRB Adopters	415	1.45	0.16	316	2.77	0.21	162	2.01	0.31	253	1.09	0.18
Combined	3,345	0.79	0.05	2,533	2.13	0.07	1,249	0.52	0.11	2,096	0.95	0.05
Diff		−0.75	0.16		−0.74	0.21		−1.72	0.32		−0.16	0.16
t (d.f.)		−4.77	(3,343)		−3.43	(2,531)		−5.29	(1,247)		−1.02	(2,094)
Ha: *Diff*<0	Prob:	0.000		Prob:	0.000		Prob:	0.000		Prob:	0.153	
Ha: *Diff*>0	Prob:	1.000		Prob:	1.000		Prob:	1.000		Prob:	0.848	

Panel 3E: The change in ESG scores (d*ESG*) – Before and after the introduction of the PRB in 2019

	PRB adopters			Non adopters		
	Obs.	Mean	Std Err	Obs.	Mean	Std Err
Pre-PRB	165	1.97	0.30	1,087	0.29	0.12
Post-PRB	250	1.11	0.18	1,843	0.93	0.05
Combined	415	1.45	0.16	2,930	0.70	0.06
Diff		0.86	0.33		−0.64	0.11
t (d.f.)		2.62	(413)		−5.626	(2,928)
Ha: *Diff*<0	Prob:	0.995		Prob:	0.000	
Ha: *Diff*>0	Prob:	0.005		Prob:	1.000	

Notes: *Diff* is the difference in means, calculated as the mean in the 1st row minus the mean in the 2nd row of each table. The dataset is a balanced panel of 1,326 banks from 29 countries over 7 years (2016—2022). The pre-PRB period is 2016–2018, and post-PRB period is 2019–2022. The variables *ESG* and *dESG* are defined in Table 1. *dESG2* is the change in ESG score over two years. The variables were winsorized at the 1st and 99th percentiles, as explained in the notes to Table 2. The null hypothesis in Tables 3B-3D is H0: Diff = 0.

The results are presented in Tables 5 and 6.

The first step is to conduct a logistic regression on the PRB adoption decision variable (*Adopter*) to match the treatment group (PRB adopters) with a comparable sample of non-adopters. The regression is performed using the full sample from the period 2016–2017 (two years before the introduction of the PRB). The explanatory variables include *EP Adopter*, *size*, *size2, ROA, tier1*, and *Reg Qlty*.

Column (1) of panel 5A presents the results for the logistic regression conducted on the full sample. Excluding *ROA* and *Reg Qlty*, the

Table 4
Assessing the impact of being a PRB adopter on ESG disclosure scores.

Panel 4A: The dependent variable is ESG score (*ESG*)

	(1)	(2)	(3)	(4)	(5)
	Entire period	**Entire period**	**Entire period**	**Entire period**	**Post PRB**
Adopter		**7.607******	**9.350******	**3.844******	**4.308******
		(1.663)	(1.679)	(1.338)	(1.284)
		0.000	*0.000*	*0.004*	*0.001*
Adopter Post		**4.031******	**1.071*****	**1.002****	
		(0.463)	(0.495)	(0.523)	
		0.000	*0.030*	*0.055*	
EP Adopter		**13.010******	**13.297******	**3.317*****	**2.945****
		(1.876)	(1.883)	(1.649)	(1.535)
		0.000	*0.000*	*0.044*	*0.055*
size	**−10.204******			**−8.444******	**−1.876**
	(1.208)			(1.301)	(1.240)
	0.000			*0.000*	*0.130*
size2	**0.651******			**0.540******	**0.251******
	(0.060)			(0.067)	(0.062)
	0.000			*0.000*	*0.000*
ROA	**−0.403****			**−0.376****	**0.215**
	(0.211)			(0.209)	(0.211)
	0.056			*0.073*	*0.309*
tier1	**0.152*****			**0.133*****	**0.062**
	(0.066)			(0.066)	(0.057)
	0.020			*0.045*	*0.281*
Reg Qlty	**−0.780****			**−1.055*****	**−1.381******
	(0.460)			(0.435)	(0.361)
	0.090			*0.015*	*0.000*
Constant	**67.937******	**32.893******	**29.603******	**61.455******	**30.990******
	(5.991)	(0.343)	(0.336)	(6.193)	(6.226)
	0.000	*0.000*	*0.000*	*0.000*	*0.000*
Year fixed effects	**Yes**	**No**	**Yes**	**Yes**	**No**
Observations	3,151	3,794	3,794	3,139	1,783
Number of banks	539	646	646	537	528
Degree Freedom	11	3	9	14	7
chi2	1028	571.2	1132	1443	1345
Prob. Chi2	0	0	0	0	0

Panel 4B: The dependent variable is the change in the ESG score, (*dESG*)

	(1)	(2)	(3)	(4)	(5)	(6)
	Entire period	**Entire period**	**Entire period**	**Entire period**	**Entire period**	**Entire period**
Lagged ESG	**−0.079******	**−0.073******			**−0.032******	**−0.084******
	(0.009)	(0.009)			(0.006)	(0.010)
	0.000	*0.000*			*0.000*	*0.000*
Adopter			**1.258******	**1.469******	**1.098******	**0.812******
			(0.330)	(0.359)	(0.244)	(0.264)
			0.000	*0.000*	*0.000*	*0.002*
Adopter Post			**−0.862*****	**−1.330******	**−0.604*****	**−0.516****
			(0.349)	(0.382)	(0.286)	(0.290)
			0.014	*0.000*	*0.035*	*0.075*
EP Adopter			**0.039**	**−0.028**	**0.235**	**0.063**
			(0.244)	(0.255)	(0.213)	(0.242)
			0.872	*0.914*	*0.270*	*0.793*
size	**0.664******	**0.762******				**0.743******
	(0.211)	(0.211)				(0.217)
	0.002	*0.000*				*0.001*
size2	**−0.015**	**−0.021****				**−0.020****
	(0.011)	(0.011)				(0.011)
	0.167	*0.057*				*0.072*
ROA	**−0.244*****	**−0.228*****				**−0.231*****
	(0.100)	(0.103)				(0.101)
	0.015	*0.027*				*0.023*
tier1	**0.062******	**0.059******				**0.056******
	(0.019)	(0.019)				(0.020)
	0.001	*0.002*				*0.005*
Reg Qlty	**−0.397******	**−0.399******				**−0.434******
	(0.099)	(0.100)				(0.102)
	0.000	*0.000*				*0.000*

(*continued on next page*)

Table 4 (*continued*)

Panel 4B: The dependent variable is the change in the ESG score, (*dESG*)

	(1)	(2)	(3)	(4)	(5)	(6)
	Entire period	**Entire period**	**Entire period**	**Entire period**	**Entire period**	**Entire period**
Constant	**−1.173**	**−1.373**	**0.695*****	**−3.861*****	**2.079*****	**−1.206**
	(1.150)	(1.139)	(0.053)	(0.486)	(0.207)	(1.187)
	0.308	*0.228*	*0.000*	*0.000*	*0.000*	*0.309*
Year fixed effects	**No**	**Yes**	**No**	**Yes**	**No**	**No**
Observations	2,623	2,623	3,332	3,332	3,140	2,613
Number of banks	519	519	622	622	621	517
Degree Freedom	6	11	3	9	4	9
chi2	141.2	173.8	24.90	175.6	36.35	167.4
Prob. Chi2	0	0	0	0	0	0

Notes: Mixed-effects regressions were performed on a balanced panel of 1,326 banks from 29 countries over 7 years, from 2016 to 2022. Variables are defined in Table 1. All non-dummy variables were winsorized at the 1st and 99th percentiles, as explained in the notes to Table 2. Robust standard errors are shown in parentheses and are adjusted for clusters by banks. P-values are in italics. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Mixed-effects regressions were performed on a balanced panel of 1,326 banks from 29 countries over 7 years, from 2016 to 2022. Variables are defined in Table 1. All non-dummy variables were winsorized at the 1st and 99th percentiles, as explained in the notes to Table 2. Robust standard errors are shown in parentheses and are adjusted for clusters by banks. P-values are in italics. *** $p < 0.01$, ** $p < 0.05$, * $p < 0$.

remaining explanatory variables are significant, including the EP adopter dummy, which enters with a positive and highly significant estimated coefficient. The pseudo-R^2 is 44 %, and the Wald-Chi^2 test is highly significant. In contrast, column (2) of panel 5A presents the results when the regression is performed on the PSM sample. Here, the coefficients lose their significance, the pseudo-R^2 drops dramatically, and the Wald-Chi^2 test is no longer significant. Consistent with Dhaliwal et al. (2016), these results suggest that our matching procedure has yielded a sample where none of the control variables can predict which banks are more likely to adopt the RBP. The similarity between the treated and untreated groups is further confirmed in panel 5B, which displays the descriptive statistics for the propensity scores, demonstrating practically no differences between the two groups.

Panel 5C compares the association between PRB adoption and EP adoption in the full sample and PSM sample. While the null hypothesis of no association is rejected at the 1 % level in the full sample, it is not rejected in the PSM sample. This provides further support for the similarity between PRB adopters and non-adopters in the PSM sample. Additionally, it is worth noting that the positive association between EP and PRB adoption in the full sample aligns with the results of the logistic regression in column (1) of panel 5A. This suggests that some banks, likely those more committed to sustainable banking practices, demonstrate this commitment by participating in emerging self-regulations as they arise.

While panel 5C tests for the association between PRB adoption and EP adoption, represented by a dummy variable, panel 5D examines differences in the means of the continuous explanatory variables between PRB adopters and non-adopters in the full and PSM samples. The results indicate significant differences between PRB adopters and non-adopters in the full sample, but these differences disappear in the PSM sample. Overall, the results of Table 5 suggest that our matching procedure has successfully yielded a sample of banks that are similar in all aspects except their decision to adopt the PRB (Appendix Table 2 provides the list of banks in the PSM sample).

With the PSM sample at hand, we can now revisit the original model, which aims to explain both the level of ESG disclosure scores and the yearly changes in scores. Table 6 presents the results of running this model on the PSM sample across both the entire and post-PRB periods. The results continue to support H1 and are consistent with those presented in Table 4. PRB signatory banks tend to have higher ESG disclosure scores and show greater commitment to increasing their scores, at least in the pre-PRB period, as do banks in weak regulatory environments.

Lastly, we conduct difference-in-differences (DiD) regressions to estimate the average effect of PRB adoption on both ESG scores and changes in those scores. Table 7 presents the results for both the full sample and the PSM sample. In columns (1) and (3), we examine the full and PSM samples, respectively, over the entire period (2016–2022), assessing the difference in ESG scores between PRB adopters in the post-adoption period and others (non-adopters and PRB adopters in the pre-adoption period). Column (2) focuses on the full sample in the post-PRB period, evaluating the difference in ESG scores between PRB adopters and non-adopters. The remaining columns follow a similar procedure, replacing *ESG* with *dESG* as the dependent variable. Overall, the results in Table 7 consistently show a positive and significant average treatment effect across all specifications, supporting H1.

6. Conclusions

ESG disclosure is argued to benefit firms in several ways. It increases transparency, thereby reducing uncertainty, increasing stakeholders' trust, and reinforcing the firm's legitimacy, reputation, credibility, and access to funds (Boiral and Heras-Saizarbitoria, 2017; Garcia et al., 2017; Li et al., 2018; Albarrak et al., 2019; Miralles-Quirós et al., 2019; Shin et al., 2023). These benefits imply that ESG disclosure may increase the firm's value (e.g., Bansal and Clelland, 2004; Orlitzky and Benjamin, 2001; Gracia et al., 2017).

In this paper, we explore the relationship between banks' commitment to self-regulation and ESG disclosure, an investigation that is important for two main reasons. First, it addresses sustainability issues within the financing sector, where banks play a crucial role in sustainable development by allocating funds to investments (Demirgüç-Kunt and Levine, 2008; Contreras et al., 2019; Bhutta et al.,

Table 5

Propensity score matching Panel 5A: Logistic regressions of the decision to adopt the RBP (*Adopter*) in the Pre-adoption period, prior to 2018. Panel 5B: Summary for propensity scores for treatment and control groups. Panel 5C: Assessing the fit of the propensity score matching (PSM) procedure by comparing the association between PRB adopters and EP (Equator Principles) adopters in the full sample and in the PSM sample. Panel 5D: Assessing the fit of the propensity score matching (PSM) procedure by comparing the means of the continuous explanatory variables of adopter and non-adopter banks in the full sample and in the PSM sample.

	(1)	(2)
	Full sample	**PSM sample**
EP Adopter	**2.107*****	**−0.051**
	(0.293)	(0.480)
	0.000	*0.915*
size	**2.397*****	**−0.004**
	(0.546)	(0.793)
	0.000	*0.996*
size2	**−0.085*****	**−0.000**
	(0.025)	(0.036)
	0.001	*0.992*
ROA	**−0.245**	**−0.053**
	(0.204)	(0.240)
	0.231	*0.825*
tier1	**0.178*****	**0.002**
	(0.028)	(0.042)
	0.000	*0.959*
Reg Qlty	**0.275**	**0.071**
	(0.188)	(0.225)
	0.143	*0.754*
Constant	**−20.260*****	**0.032**
	(3.145)	(4.379)
	0.000	*0.994*
Observations	1,264	166
Degree Freedom	6	6
chi2	203.7	0.271
Prob. Chi2	0	1.000
Pseudo R2	0.435	0.00118

Adopter	N	Mean	SD	Min	p25	p50	p75	Max
0	83	0.303	0.278	0.002	0.096	0.191	0.444	0.926
1	83	0.301	0.276	0.002	0.096	0.194	0.442	0.921
Difference:		0.001	0.002	0.000	0.000	−0.003	0.002	0.005
Total:	166	0.302	0.276	0.002	0.096	0.193	0.442	0.926

Full sample				**PSM sample**			
	PRB Adopter				**PRB Adopter**		
EP Adopter	**0**	**1**	**Total**	**EP Adopter**	**0**	**1**	**Total**
0	2,434	84	**2,518**	**0**	64	65	**129**
	(2,357.8)	(160.2)	**(2,518.00)**		(64.5)	(64.5)	**(129.0)**
1	38	84	**122**	**1**	19	18	**37**
	(114.2)	(7.8)	**(122.0)**		(18.5)	(18.5)	**(37.0)**
Total	2,472	168	**2,640**	**Total**	**83**	**83**	**166**
	(2,472.00)	(168.0)	**(2,640.00)**		**(83.0)**	**(83.0)**	**(166.0)**
Pearson chi2(1)	838.2293	Prob.	0.000	Pearson chi2(1)	0.0348	Prob.	0.852

	Full sample			PSM sample		
	Mean Non-adopters	**Mean Adopters**	**Difference**	**Mean Non-adopters**	**Mean Adopters**	**Difference**
size	8.607	11.785	−3.178***	11.067	11.011	0.056
size2	78.122	143.550	−65.428***	126.523	125.255	1.268
ROA	0.846	0.726	0.120**	0.786	0.759	0.027
tier1	13.644	15.402	−1.758***	15.076	15.207	−0.131
Reg Qlty	1.170	1.019	0.151***	0.998	1.049	−0.051

Notes: Logistic regressions were performed on a balanced panel of 1,326 banks from 29 countries over 2 years, from 2016 to 2017. Variables are defined in Table 1. All non-dummy variables were winsorized at the 1st and 99th percentiles, as explained in the notes to Table 2. The matching procedure included a 1-to-1 matching without replacement of nearest neighbor, controlling for maximum distance with a caliper of 0.01. Robust standard errors are shown in parentheses and are adjusted for clusters by banks. P-values are in italics. ***, **, and * denote significance at 1 %, 5 %, and 10 %, respectively.

Notes: A two-sample *t* test (assuming equal variances) was performed to compare the means of the explanatory variables in the pre-adoption period (prior to 2018) across the treated (PRB Adopters) and untreated (non-adopters) groups. Variables are defined in Table 1. All non-dummy variables were winsorized at the 1st and 99th percentiles, as explained in the notes to Table 2. H0: difference = 0; Ha: Difference ≠ 0. ***, **, and * denote significance at 1 %, 5 %, and 10 %, respectively.

Table 6
Using the PSM sample to assess the impact of being a PRB adopter on ESG disclosure scores.

	(1)	(2)	(3)	(4)	(5)	(6)
	ESG	*ESG*	*ESG*	*dESG*	*dESG*	*dESG*
	PSM sample	PSM sample	PSM sample	PSM sample	PSM sample	PSM sample
	Entire period	**Post PRB**	**Post PRB**	**Entire period**	**Post PRB**	**Post PRB**
Lagged ESG				**−0.081*****	**−0.110*****	**−0.107*****
				(0.014)	(0.020)	(0.019)
				0.000	*0.000*	*0.000*
Adopter	**2.693**	**4.928*****	**5.344*****	**0.707****	**0.703****	**0.743****
	(1.695)	(1.647)	(1.599)	(0.330)	(0.352)	(0.350)
	0.112	*0.003*	*0.001*	0.032	0.046	0.034
Adopter Post	**3.775*****			**−0.217**		
	(0.696)			(0.408)		
	0.000			*0.594*		
EP Adopter	**1.670**	**6.473*****	**5.040****	**0.375**	**0.436**	**0.416**
	(2.660)	(2.211)	(2.287)	(0.298)	(0.458)	(0.452)
	0.530	*0.003*	*0.028*	*0.209*	*0.341*	*0.358*
size	**2.334**	**3.977**	**8.551***	**1.630****	**2.555****	**2.656*****
	(4.325)	(5.057)	(4.883)	(0.679)	(1.032)	(1.021)
	0.590	*0.432*	*0.080*	*0.016*	*0.013*	*0.009*
size2	**0.124**	**−0.025**	**−0.195**	**−0.061****	**−0.096****	**−0.100****
	(0.198)	(0.223)	(0.218)	(0.029)	(0.044)	(0.043)
	0.533	*0.909*	*0.371*	*0.039*	*0.027*	*0.021*
ROA	**0.123**	**−1.121****	**−0.382**	**−0.205**	**−0.421****	**−0.268**
	(0.378)	(0.500)	(0.410)	(0.148)	(0.174)	(0.168)
	0.746	*0.025*	*0.352*	*0.165*	*0.015*	*0.110*
tier1	**0.171***	**0.138**	**0.160**	**0.018**	**0.000**	**−0.005**
	(0.100)	(0.123)	(0.129)	(0.029)	(0.041)	(0.040)
	0.088	*0.262*	*0.216*	*0.547*	*0.993*	*0.895*
Reg Qlty	**−1.550***	**−1.485***	**−2.580*****	**−0.382****	**−0.474****	**−0.440****
	(0.879)	(0.831)	(0.834)	(0.165)	(0.205)	(0.209)
	0.078	*0.074*	*0.002*	*0.020*	*0.021*	*0.035*
Constant	**−2.662**	**−0.630**	**−29.089**	**−5.595**	**−9.675**	**−10.570***
	(23.243)	(27.733)	(26.718)	(3.808)	(5.968)	(5.921)
	0.909	*0.982*	*0.276*	*0.142*	*0.105*	*0.074*
Year fixed effects	**No**	**Yes**	**No**	**No**	**Yes**	**No**
Observations	644	351	351	534	347	347
Number of banks	110	107	107	107	106	106
Degree Freedom	8	10	7	9	11	8
chi2	371.0	301.4	297.4	53.61	61.44	51.34
Prob. Chi2	0	0	0	0	0	0

Notes: Mixed-effects regressions were performed on a balanced panel of 1,326 banks from 29 countries over the period 2016 to 2022. Variables are defined in Table 1. All non-dummy variables were winsorized at the 1st and 99th percentiles, as explained in the notes to Table 2. Robust standard errors are shown in parentheses and are adjusted for clusters by banks. P-values are in italics. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

2022; Grijavo and García-Wang, 2023; Úbeda et al., 2023). Second, given the heavily regulated nature of the banking industry, our study contributes to the ongoing debate about the balance between self- and government-imposed regulation.

The self-regulation initiative examined in this study is the PRB, a set of six principles for responsible banking that was launched in 2019 to guide signatories in sustainable development across *all* areas of banking. By investigating whether banks committed to the PRB exhibit higher ESG disclosure scores, we contribute to the debate regarding the effectiveness of self-regulation (e.g., Pitkänen et al., 2016; Folke et al., 2019; Kolcava et al., 2021). Specifically, we explore whether self-regulation signals ESG-conscious management or merely represents a form of greenwashing (Mahoney et al., 2013; Finger et al., 2018; Caby et al., 2020; Uyar et al, 2020).

Our findings indicate that self-regulation in sustainable banking is positively associated with higher ESG disclosure scores. This relationship remains consistent whether self-regulation is represented by the comprehensive PRB, which is the focus of this study, or by an earlier self-regulation scheme, the EP, which serves as a control variable. This aligns with self-regulation theory (Lyon and Maxwell, 2003), which suggests that corporations adopt self-regulation to preemptively address stricter future government regulations. It also supports signaling theory, which asserts that self-regulation reflects responsible conduct (Scholtens and Dam, 2007). Therefore, we conclude that adopting self-regulation in responsible banking serves as a credible signal of a bank's commitment to sustainability. Furthermore, our PSM logistic regression results suggest that early adopters of self-regulation (the EP) are more likely to engage in subsequent initiatives (the PRB), highlighting their ongoing dedication to responsible banking practices.

One caveat is worth noting. Our results show that following the PRB adoption period, differences in ESG *improvements* between PRB

Table 7
Difference-in-difference regressions of ESG disclosure score.

	(1)	(2)	(3)	(4)	(5)	(6)
	ESG	ESG	ESG	dESG1	dESG1	dESG1
	Full sample	**Full sample**	**PSM sample**	**Full sample**	**Full sample**	**PSM sample**
	Entire period	**Post PRB**	**Entire period**	**Entire period**	**Post PRB**	**Entire period**
ATET						
Adopter Post	2.717***	0.796***	3.432***	0.365	0.661***	0.826*
(1 vs 0)	(0.506)	(0.234)	(0.722)	(0.313)	(0.204)	(0.476)
	0.000	0.001	0.000	0.245	0.001	0.086
Controls						
Lagged ESG				**−0.403*****	**−0.538*****	**−0.436*****
				(0.032)	(0.037)	(0.059)
				0.000	*0.000*	*0.000*
size	**−7.974*****	**−6.610****	**−9.675**	**−5.384*****	**−4.374****	**−7.801**
	(2.168)	(3.052)	(6.117)	(1.471)	(1.845)	(5.252)
	0.000	*0.031*	*0.117*	*0.000*	*0.018*	*0.140*
size2	**0.765*****	**0.623*****	**0.800****	**0.377*****	**0.356*****	**0.506****
	(0.130)	(0.174)	(0.305)	(0.087)	(0.107)	(0.249)
	0.000	*0.000*	*0.010*	*0.000*	*0.001*	*0.044*
ROA	**0.084**	**0.277**	**0.375**	**−0.100**	**−0.007**	**−0.109**
	(0.213)	(0.212)	(0.385)	(0.128)	(0.153)	(0.251)
	0.695	*0.193*	*0.332*	*0.435*	*0.962*	*0.666*
tier1	**0.220****	**0.076**	**0.132**	**0.071**	**0.040**	**0.055**
	(0.088)	(0.067)	(0.113)	(0.052)	(0.051)	(0.075)
	0.012	*0.254*	*0.244*	*0.171*	*0.437*	*0.462*
Reg Qlty	**0.500**	**−0.340**	**−1.111**	**−0.680***	**−0.538**	**−0.823**
	(0.346)	(0.546)	(1.426)	(0.369)	(0.383)	(1.067)
	0.149	*0.533*	*0.437*	*0.066*	*0.161*	*0.442*
Constant	**35.205*****	**39.550*****	**45.895**	**30.654*****	**28.220*****	**41.656**
	(8.833)	(13.768)	(30.628)	(6.426)	(8.365)	(28.079)
	0.000	*0.004*	*0.137*	*0.000*	*0.001*	*0.141*
Observations	3,151	1,789	644	2,623	1,744	534
Number of banks	539	530	110	519	517	107
Degree Freedom	538	529	109	518	516	106

Notes: ATET estimates were adjusted for covariates and panel effects using 1,326 banks from 29 countries over the period from 2016 to 2022. Variables are defined in Table 1. All non-dummy variables were winsorized at the 1st and 99th percentiles, as explained in the notes to Table 2. Robust standard errors are shown in parentheses and are adjusted for clusters by banks. P-values are in italics. ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

adopters and non-adopters nearly disappear. The growing focus on sustainability (Gupta and Kashiramka, 2024) seems to have prompted non-adopters to increase their engagement in ESG as well, bringing their scores closer to those of PRB adopters. A possible explanation is that PRB adopters, as industry leaders in responsible banking, drive the broader adoption of sustainability practices (Johannsdottir and McInerney, 2018). From this perspective, early adopters play a crucial role by setting an example and creating peer pressure that accelerates the speed and scale of industry-wide implementation (Contreras et al., 2019). A more skeptical explanation is that banks with improved ESG scores may adopt the PRB primarily to minimize compliance burdens, potentially diminishing their commitment to responsible banking after adoption. This raises questions about the long-term effectiveness of self-regulation.[15]

The results of this paper further show that a weak regulatory environment is associated with higher ESG disclosure. This aligns with the notion that weak institutions create uncertainty and risk, prompting firms to enhance their disclosure, reduce information asymmetries, and establish trust by demonstrating responsible conduct (Egginton and McBrayer, 2019; Huang et al., 2022; Úbeda et al., 2022b). Additionally, we find that large banks tend to have higher ESG disclosure scores, although this relationship is nonlinear: ESG scores initially decrease with bank size but then increase as banks grow larger. This reflects the greater visibility of very large banks, which attracts stakeholder attention and encourages transparency (Giannarakis, 2014; Gianfrate 2017; Tamimi and Sebastianelli, 2017). Taken together, our results shed light on the value of self-regulation as a signal of genuine commitment to responsible banking and on how market forces address regulatory voids.

CRediT authorship contribution statement

Ronny Manos: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Maya Finger:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haim Boukai:** Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A.1
List of countries.

Country	Number of banks	Yearly average regulatory quality (Reg Qlty) over the period 2016–2022
AUSTRALIA	12	1.86
BAHRAIN	9	0.60
BANGLADESH	35	−0.86
BELGIUM	6	1.30
BRAZIL	18	−0.15
BRITAIN	10	1.67
CHINA	60	−0.26
COLOMBIA	8	0.30
DENMARK	18	1.68
FINLAND	4	1.84
FRANCE	16	1.20
GERMANY	5	1.72
GREECE	6	0.39
ICELAND	3	1.39
IRELAND	3	1.63
ITALY	33	0.69
JAPAN	89	1.34
MEXICO	7	0.08
MOROCCO	6	−0.13
NETHERLANDS	3	1.89
NIGERIA	13	−0.92
NORWAY	39	1.72
SOUTH AFRICA	7	0.07
SOUTH KOREA	15	1.09
SPAIN	7	0.90
SWEDEN	4	1.79
SWITZERLAND	48	1.76
TURKEY	13	0.07
UNITED STATES	829	1.43

Notes: Regulatory Quality reflects perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development. The score ranges from −2.5 (weak) to 2.5 (strong) governance performance (Kaufmann et. al., 2011).

Table A.2
List of banks in the PSM sample.

Bank Name	Country	Propensity Score	Adopter (Treated)
AUST AND NZ BANKING GROUP	AUSTRALIA	0.782	1
MACQUARIE GROUP LTD	AUSTRALIA	0.225	0
NATIONAL AUSTRALIA BANK LTD	AUSTRALIA	0.788	1
NATIONAL AUSTRALIA BANK LTD	AUSTRALIA	0.775	1
AHLI UNITED BANK B.S.C	BAHRAIN	0.502	1
AL BARAKA GROUP B.S.C	BAHRAIN	0.031	0
AL BARAKA GROUP B.S.C	BAHRAIN	0.144	0
BNP PARIBAS FORTIS SA- AUC	BELGIUM	0.293	0
BNP PARIBAS FORTIS SA- AUC	BELGIUM	0.363	0
KBC GROUP NV	BELGIUM	0.861	1
BANCO ABC BRASIL SA	BRAZIL	0.031	0
BANCO DA AMAZONIA SA	BRAZIL	0.027	1

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Table A.2 (*continued*)

Bank Name	Country	Propensity Score	Adopter (Treated)
BANCO DA AMAZONIA SA	BRAZIL	0.028	1
BANCO DO BRASIL S.A.	BRAZIL	0.695	0
BARCLAYS PLC	BRITAIN	0.876	1
BARCLAYS PLC	BRITAIN	0.908	1
HSBC HOLDINGS PLC	BRITAIN	0.884	0
HSBC HOLDINGS PLC	BRITAIN	0.895	0
LLOYDS BANKING GROUP PLC	BRITAIN	0.896	1
NATIONWIDE BUILDING SOC-CCDS	BRITAIN	0.868	0
NATIONWIDE BUILDING SOC-CCDS	BRITAIN	0.913	0
NATWEST GROUP PLC	BRITAIN	0.926	0
NATWEST GROUP PLC	BRITAIN	0.912	0
VIRGIN MONEY UK PLC	BRITAIN	0.270	0
AGRICULTURAL BANK OF CHINA-H	CHINA	0.143	0
BANK OF BEIJING CO LTD −A	CHINA	0.114	0
BANK OF CHONGQING CO LTD-H	CHINA	0.324	0
BANK OF CHONGQING CO LTD-H	CHINA	0.305	0
BANK OF JIANGSU CO LTD-A	CHINA	0.526	0
BANK OF JIUJIANG CO LTD-H	CHINA	0.043	0
BANK OF QINGDAO CO LTD-H	CHINA	0.089	0
BANK OF SHANGHAI CO LTD-A	CHINA	0.125	0
CHINA EVERBRIGHT BANK CO-A	CHINA	0.139	0
CHINA MERCHANTS BANK-A	CHINA	0.191	0
CHINA ZHESHANG BANK CO LTD-H	CHINA	0.106	0
HUAXIA BANK CO LTD-A	CHINA	0.106	1
HUAXIA BANK CO LTD-A	CHINA	0.106	1
IND & COMM BK OF CHINA-A	CHINA	0.195	1
IND & COMM BK OF CHINA-A	CHINA	0.194	1
INDUSTRIAL BANK CO LTD −A	CHINA	0.531	1
PING AN BANK CO LTD-A	CHINA	0.106	0
QINGDAO RURAL COMMERCIAL B-A	CHINA	0.053	0
BANCOLOMBIA SA	COLOMBIA	0.337	1
JYSKE BANK-REG	DENMARK	0.369	1
JYSKE BANK-REG	DENMARK	0.388	1
ALANDSBANKEN-A	FINLAND	0.031	1
ALANDSBANKEN-A	FINLAND	0.044	1
OMA SAASTOPANKKI OYJ	FINLAND	0.047	0
BNP PARIBAS	FRANCE	0.769	1
BNP PARIBAS	FRANCE	0.775	1
COMMERZBANK AG	GERMANY	0.436	1
DEUTSCHE PFANDBRIEFBANK AG	GERMANY	0.494	0
DEUTSCHE PFANDBRIEFBANK AG	GERMANY	0.364	0
ALPHA SERVICES AND HOLDINGS	GREECE	0.272	1
ALPHA SERVICES AND HOLDINGS	GREECE	0.310	1
EUROBANK ERGASIAS SERVICES A	GREECE	0.279	1
EUROBANK ERGASIAS SERVICES A	GREECE	0.287	1
NATIONAL BANK OF GREECE	GREECE	0.336	0
NATIONAL BANK OF GREECE	GREECE	0.301	0
PIRAEUS FINANCIAL HOLDINGS S	GREECE	0.292	1
PIRAEUS FINANCIAL HOLDINGS S	GREECE	0.224	1
ARION BANKI HF	ICELAND	0.287	1
ARION BANKI HF	ICELAND	0.253	1
AIB GROUP PLC	IRELAND	0.442	1
AIB GROUP PLC	IRELAND	0.516	1
BANK OF IRELAND GROUP PLC	IRELAND	0.332	1
BANK OF IRELAND GROUP PLC	IRELAND	0.397	1
BANCA MEDIOLANUM SPA	ITALY	0.404	0
BANCA MONTE DEI PASCHI SIENA	ITALY	0.145	1
BANCA MONTE DEI PASCHI SIENA	ITALY	0.358	1
BPER BANCA	ITALY	0.198	0
CREDITO EMILIANO SPA	ITALY	0.147	0
MEDIOBANCA SPA	ITALY	0.169	0
UNICREDIT SPA	ITALY	0.700	1
HACHIJUNI BANK LTD/THE	JAPAN	0.444	0
HOKKOKU FINANCIAL HOLDINGS I	JAPAN	0.128	0
SHIGA BANK LTD/THE	JAPAN	0.252	0
GRUPO FINANCIERO BANORTE-O	MEXICO	0.097	1
GRUPO FINANCIERO BANORTE-O	MEXICO	0.139	1
GRUPO FINANCIERO INBURSA-O	MEXICO	0.096	0
ABN AMRO BANK NV-CVA	NETHERLANDS	0.907	1
GUARANTY TRUST HOLDING CO PL	NIGERIA	0.097	0

(*continued on next page*)

Table A.2 (*continued*)

Bank Name	Country	Propensity Score	Adopter (Treated)
WEMA BANK PLC	NIGERIA	0.002	1
WEMA BANK PLC	NIGERIA	0.003	1
ZENITH BANK PLC	NIGERIA	0.125	1
SANDNES SPAREBANK	NORWAY	0.043	1
SANDNES SPAREBANK	NORWAY	0.050	1
SBANKEN ASA	NORWAY	0.076	1
SBANKEN ASA	NORWAY	0.096	1
SPAREBANK 1 NORD-NORGE	NORWAY	0.086	1
SPAREBANK 1 OESTLANDET	NORWAY	0.121	1
SPAREBANK 1 OESTLANDET	NORWAY	0.136	1
SPAREBANK 1 SMN	NORWAY	0.097	1
SPAREBANK 1 SMN	NORWAY	0.142	1
SPAREBANK 1 SOROST-NORGE	NORWAY	0.047	1
SPAREBANK 1 SOROST-NORGE	NORWAY	0.053	1
SPAREBANK 1 SR BANK ASA	NORWAY	0.143	1
SPAREBANK 1 SR BANK ASA	NORWAY	0.166	1
SPAREBANKEN MORE-CAP CERT	NORWAY	0.079	1
SPAREBANKEN MORE-CAP CERT	NORWAY	0.088	1
SPAREBANKEN SOR	NORWAY	0.102	1
SPAREBANKEN SOR	NORWAY	0.128	1
SPAREBANKEN VEST	NORWAY	0.114	1
SPAREBANKEN VEST	NORWAY	0.170	1
TOTENS SPAREBANK	NORWAY	0.021	0
ABSA GROUP LTD	SOUTH AFRICA	0.535	1
NEDBANK GROUP LTD	SOUTH AFRICA	0.524	0
BNK FINANCIAL GROUP INC	SOUTH KOREA	0.128	0
DGB FINANCIAL GROUP INC	SOUTH KOREA	0.096	1
DGB FINANCIAL GROUP INC	SOUTH KOREA	0.104	1
INDUSTRIAL BANK OF KOREA	SOUTH KOREA	0.626	0
INDUSTRIAL BANK OF KOREA	SOUTH KOREA	0.675	0
KB FINANCIAL GROUP INC	SOUTH KOREA	0.300	1
KB FINANCIAL GROUP INC	SOUTH KOREA	0.310	1
SHINHAN FINANCIAL GROUP LTD	SOUTH KOREA	0.737	1
BANCO DE SABADELL SA	SPAIN	0.675	1
BANKIA SA	SPAIN	0.324	1
BANKINTER SA	SPAIN	0.538	0
BANKINTER SA	SPAIN	0.589	0
BANQUE CANTONALE VAUDOIS-REG	SWITZERLAND	0.279	0
BANQUE CANTONALE VAUDOIS-REG	SWITZERLAND	0.281	0
BASLER KANTONALBK – PC	SWITZERLAND	0.307	0
BASLER KANTONALBK – PC	SWITZERLAND	0.333	0
CREDIT SUISSE GROUP AG-REG	SWITZERLAND	0.921	1
JULIUS BAER GROUP LTD	SWITZERLAND	0.350	1
UBS GROUP AG-REG	SWITZERLAND	0.623	1
UBS GROUP AG-REG	SWITZERLAND	0.586	1
SEKERBANK	TURKEY	0.030	1
TURKIYE GARANTI BANKASI	TURKEY	0.127	1
TURKIYE GARANTI BANKASI	TURKEY	0.146	1
TURKIYE SINAI KALKINMA BANK	TURKEY	0.026	1
TURKIYE SINAI KALKINMA BANK	TURKEY	0.021	1
AMALGAMATED FINANCIAL CORP	UNITED STATES	0.022	1
ATLANTIC UNION BANKSHARES CO	UNITED STATES	0.030	1
ATLANTIC UNION BANKSHARES CO	UNITED STATES	0.031	1
BANK OF AMERICA CORP	UNITED STATES	0.778	0
BANK OF AMERICA CORP	UNITED STATES	0.783	0
BANNER CORPORATION	UNITED STATES	0.313	0
CAPITOL FEDERAL FINANCIAL IN	UNITED STATES	0.436	0
CAPITOL FEDERAL FINANCIAL IN	UNITED STATES	0.392	0
EAST WEST BANCORP INC	UNITED STATES	0.086	0
FARMERS & MERCHANTS BANK/CA	UNITED STATES	0.096	0
FIRST CITIZENS BCSHS –CL A	UNITED STATES	0.104	0
FIRST FOUNDATION INC	UNITED STATES	0.022	0
FIRST HAWAIIAN INC	UNITED STATES	0.080	0
FIRST INTERSTATE BANCSYS-A	UNITED STATES	0.050	0
FIRST MERCHANTS CORP	UNITED STATES	0.030	0
FIRST OF LONG ISLAND CORP	UNITED STATES	0.026	0
FLUSHING FINANCIAL CORP	UNITED STATES	0.044	0
HARBORONE BANCORP INC	UNITED STATES	0.027	0
HUNTINGTON BANCSHARES INC	UNITED STATES	0.144	0
JPMORGAN CHASE & CO	UNITED STATES	0.784	0

(*continued on next page*)

Table A.2 (*continued*)

Bank Name	Country	Propensity Score	Adopter (Treated)
JPMORGAN CHASE & CO	UNITED STATES	0.791	0
KEARNY FINANCIAL CORP/MD	UNITED STATES	0.349	0
PACIFIC PREMIER BANCORP INC	UNITED STATES	0.030	0
PLUMAS BANCORP	UNITED STATES	0.003	0
PROSPERITY BANCSHARES INC	UNITED STATES	0.102	0
PROSPERITY BANCSHARES INC	UNITED STATES	0.121	0
REGIONS FINANCIAL CORP	UNITED STATES	0.166	0
SIGNATURE BANK	UNITED STATES	0.097	0
SOUND FINANCIAL BANCORP INC	UNITED STATES	0.002	0
SVB FINANCIAL GROUP	UNITED STATES	0.135	0
UNITED BANKSHARES INC	UNITED STATES	0.076	0
VERITEX HOLDINGS INC	UNITED STATES	0.027	0
WELLS FARGO & CO	UNITED STATES	0.739	0
WELLS FARGO & CO	UNITED STATES	0.794	0

Data availability

Data will be made available on request.

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